

CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Exp-04: Binary Search with Big-O Analysis

Aim:

To search for an element in a sorted array using Binary Search and analyze its time and space complexity.

Algorithm:

1. Read the size of the sorted array.
2. Read the sorted array elements.
3. Read the element to be searched.
4. Set $low = 0$ and $high = n - 1$.
5. Find the middle element.
6. If the middle element equals the search element, display its position.
7. If the search element is greater, search the right half.
8. Otherwise, search the left half.
9. Repeat until the element is found or the search range becomes empty.
10. Display the complexity.

Source Code:

```
n = int(input("Enter number of elements: "))
arr = list(map(int, input("Enter sorted array elements: ").split()))
key = int(input("Enter element to search: "))
low = 0
high = n - 1
found = False
while low <= high:
```

```
mid = (low + high) // 2
if arr[mid] == key:
    print("Element found at position:", mid + 1)
    found = True
    break
elif arr[mid] < key:
    low = mid + 1
else:
    high = mid - 1
if not found:
    print("Element not found")
print("Time Complexity: O(log n)")
print("Space Complexity: O(1)")
```

Output:

```
Enter number of elements: 5
Enter sorted array elements: 10 20 30 40 50
Enter element to search: 40
Element found at position: 4
Time Complexity: O(log n)
Space Complexity: O(1)
```

Result:

The given element was successfully searched using Binary Search and its Big-O complexity was analyzed.